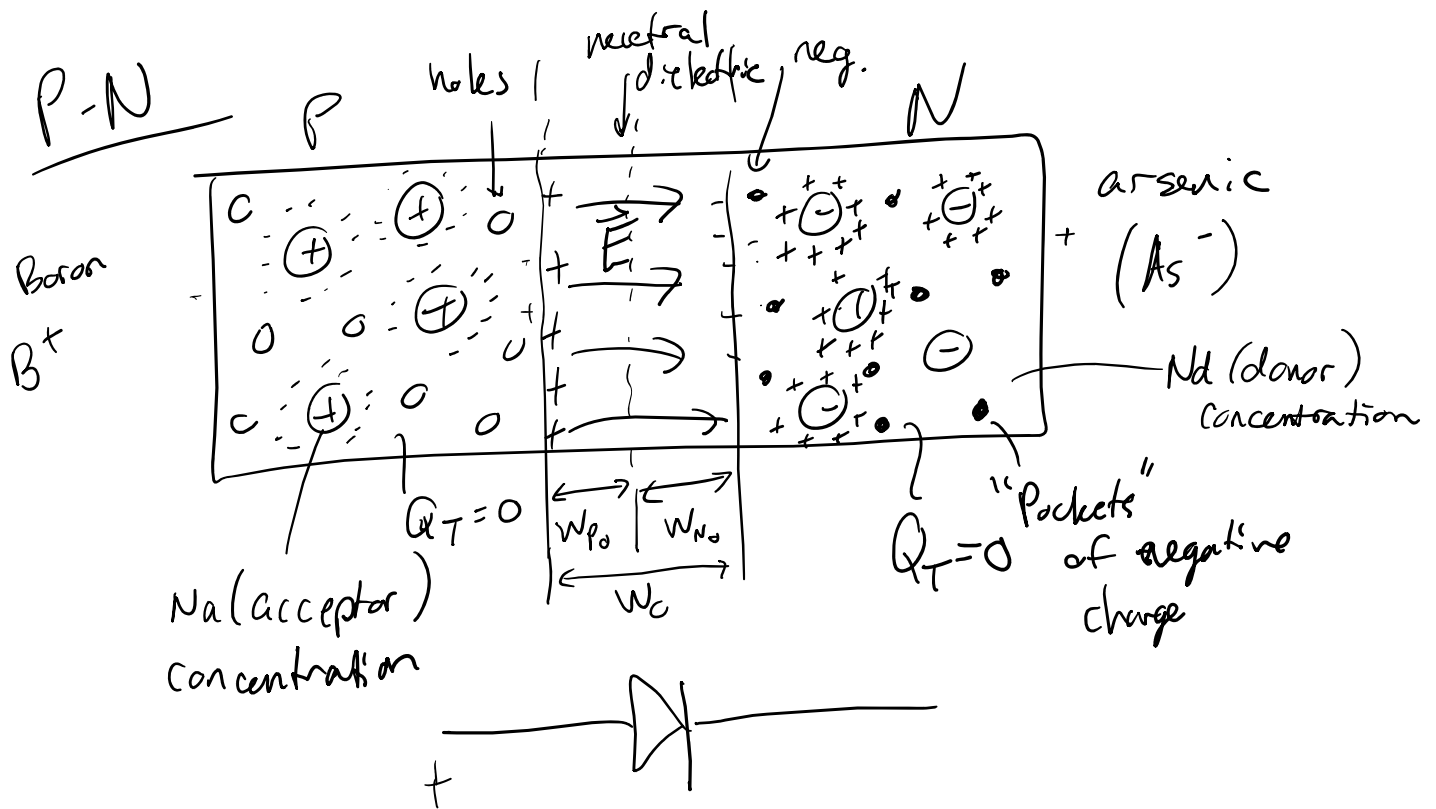




• W_0 as the depletion Region

$$W_0 = W_{p0} + W_{n0}$$

$$N_a(W_{p0}) = N_d(W_{n0})$$

$$N_a(W_0 - W_{n0}) = N_d(W_{n0})$$

$$W_0 = \frac{N_d}{N_a}(W_{n0}) + (W_{n0})$$

$$W_0 = \left(\frac{N_d}{N_a} + 1 \right) W_{n0}$$

$$W_0 = \left(\frac{N_a}{N_d} + 1 \right) W_{p0}$$

$$\frac{W_{p0}}{W_{n0}} = \frac{N_d}{N_a}$$

$$\vec{E} = -\nabla V \Rightarrow \text{Derive}$$

$$\vec{E}_0 = -\frac{q(N_d W_{n0})}{\epsilon} [uw]$$

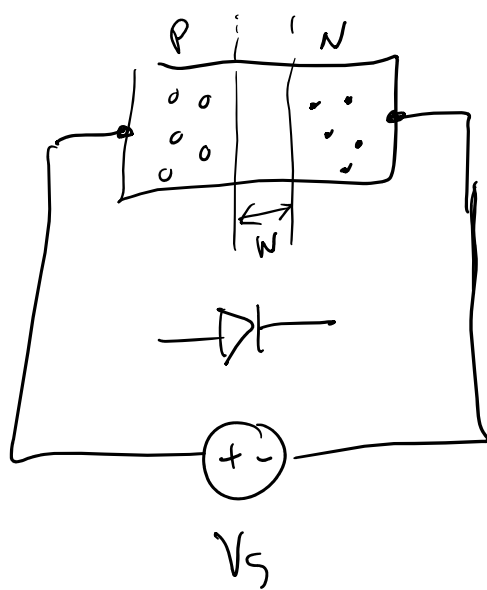
$$V_0 = \frac{1}{2} E_0 W_0 = \frac{q N_a N_d (W_0)^2}{2 \epsilon (N_a + N_d)}$$

$$\epsilon = \epsilon_r \epsilon_0 \left[\frac{F}{m} \right]$$

$$V_0 = \frac{k_B T}{q} \ln \left(\frac{N_a N_d}{(N_0)^2} \right)$$

* N_0 is initial concentration of electrons before doping
Page 1

$$k_B = 1.38 \times 10^{-23} \text{ J/Kelvin}$$

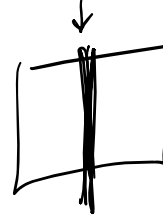


Forward Bias

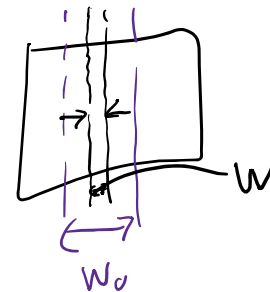
$$W = (W_0 - \Delta W_0)$$

$$V_s > 0.6 \rightarrow$$

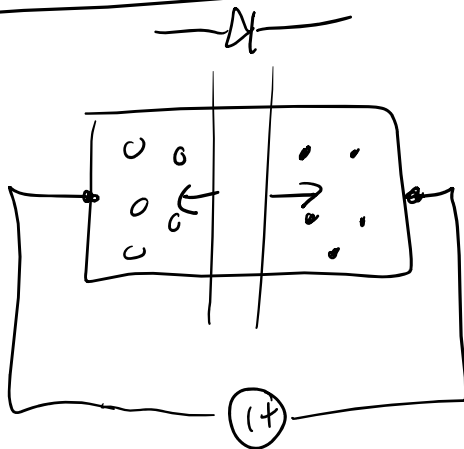
like "plates touching"



$$V_s < 0.6$$



Reverse Bias

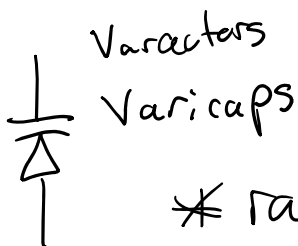


$$W = W_0 + \Delta W_0$$

* imagine the depletion region is a virtual capacitor plate.

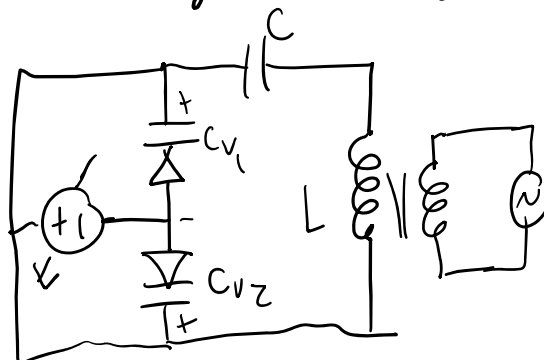
* this is a heel capacitance

* variable capacitors that are electrically controlled.



Varactors
Varicaps

* radio tuning
* tunable filter

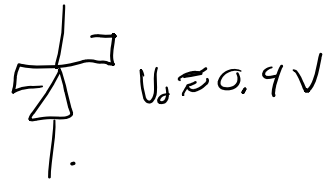
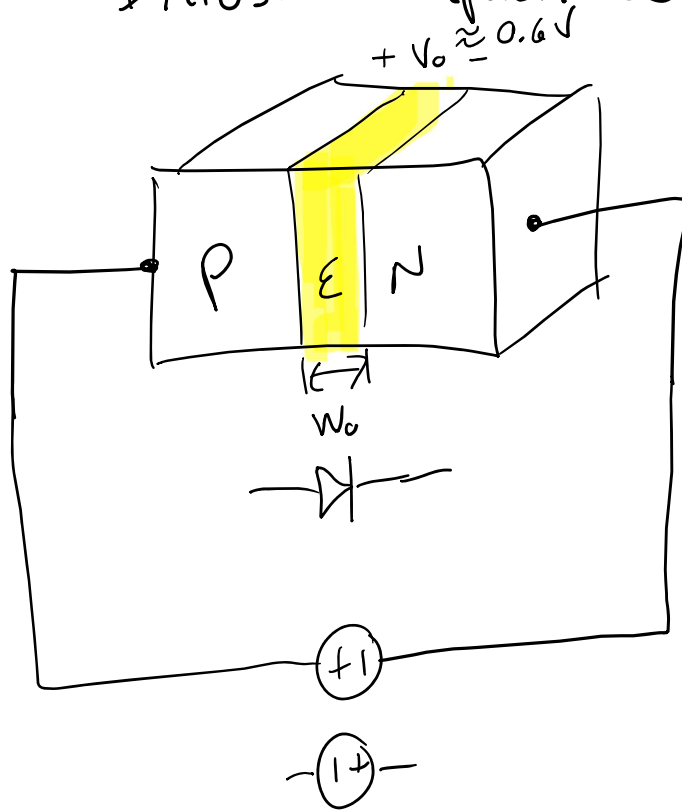


$$f = \frac{1}{2\pi\sqrt{L C_T}}$$

$C_T = \text{total capacitance}$
 $C_T = \left(\frac{1}{C_{v1}} + \frac{1}{C_{v2}} + \frac{1}{C} \right)^{-1}$

PN Junctions Exhibit 2 types of capacitance

- Depletion Region capacitance
- Diffusion Capacitance



$$C_{dep} = \epsilon \frac{S}{w_0} \quad (nF \rightarrow pF)$$

$$V_s \pm V_0 = \frac{q N_a N_d (w_0)^2}{2 \epsilon (N_a + N_d)}$$

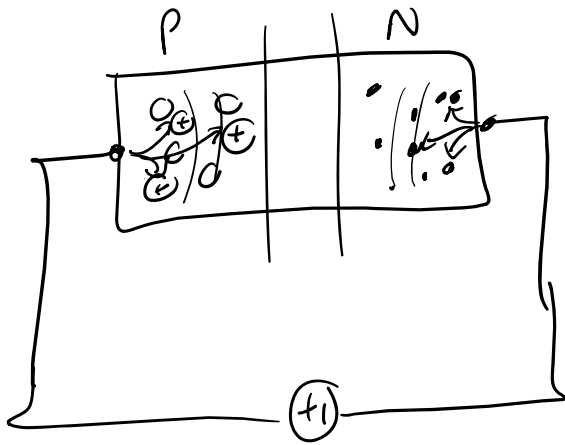
$$w_0 = \sqrt{\frac{2 \epsilon (V_0 \pm V_s) (N_a + N_d)}{q N_a N_d}}$$

$$C_{dep} = \epsilon S \left[\frac{(q N_a N_d)}{2 \epsilon (V_0 \pm V_s) (N_a + N_d)} \right]^{1/2}$$

$$C_{dep} = \frac{S}{(V_0 \pm V_s)^{1/2}} \left(\frac{\epsilon^2 q N_a N_d}{2 \epsilon (N_a + N_d)} \right)^{1/2}$$

$$C_{dep} = \frac{S}{(V_0 \pm V_s)^{1/2}} \left(\frac{q \epsilon N_a N_d}{2 (N_a + N_d)} \right)^{1/2}$$

Diffusion Capacitance



$$C_{diff} = \frac{\tau_h I}{\left(\frac{k_B T}{q}\right)} = q \frac{\tau_h I}{k_B T}$$

τ_h = time constant it takes
holes to be filled in the
entire diode.

$$C_{diff} \approx nF$$